

Lecture 06 - Rasterization

This lecture explains how continuous primitives become discrete fragments on a grid. Rasterization is the bridge from geometric descriptions to sample-based image generation.

Source chunk	course_text_parts/03_lectures/06_rasterization.txt
Extracted slide pages	61
Reader purpose	Commentary and causal explanation for human reading

How to read this lecture

- Read the section commentary before looking at the raw slide bullets.
- Use the mental model as the short version you should remember.
- Answer the check question without notes before continuing.

Slide walkthrough

Each row gives a short reading comment for one extracted slide page. Use it to turn sparse slide bullets into a human-readable path through the lecture.

Page	Slide title	How to read it
1	Untitled slide	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
2	Rasterization 1/2	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
3	Rasterization 2/2	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
4	6.1 Line Rasterization	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
5	6.1 Line Rasterization	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
6	Line Rasterization	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
7	Quality Criteria	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
8	Naïve Algorithm 1/2	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.

Page	Slide title	How to read it
9	Naïve Algorithm 2/2	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
10	Digital Difference Analyzer (DDA) 1/3	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
11	Digital Difference Analyzer (DDA) 2/3	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
12	Digital Difference Analyzer (DDA) 3/3	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
13	Midpoint-Line Algorithm 1/2	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
14	Midpoint-Line Algorithm 2/2	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
15	Implicit Function	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
16	Decision Variable ■	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
17	Update of ■ for ■	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
18	Update of ■ for ■■	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
19	Proceeding	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
20	Initializing ■	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
21	Using ■	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
22	Pseudo Code – Simple Version	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
23	Example	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.

Page	Slide title	How to read it
24	Evaluation	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
25	Pseudo Code - Reversible	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
26	6.2 Triangle Edge Rasterization	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
27	Triangle Edges 1/2	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
28	Triangle Edges 2/2	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
29	Edge Rasterization 1/3	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
30	Edge Rasterization 2/3	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
31	Edge Rasterization 3/3	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
32	6.3 Region Filling	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
33	Region Types 1/2	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
34	Region Types 2/2	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
35	Region Spezifikation	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
36	Flood Fill Algorithm	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
37	Boundary Fill Algorithm	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
38	Fill Algorithms	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?

Page	Slide title	How to read it
39	6.4 Scanline-Based Triangle Rasterization	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
40	Triangle Rasterization	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
41	Scanline-Based Rasterization	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
42	Algorithm Overview	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
43	Slope Calculation	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
44	Pseudo Code	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
45	Segment Rasterization	Read this as the main data-flow story: scene/model data is processed step by step until only valid framebuffer updates remain.
46	Vertex Attribute Interpolation	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
47	Incremental Interpolation 1/3	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
48	Incremental Interpolation 2/3	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
49	Incremental Interpolation 3/3	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
50	6.5 Tile-Based Triangle Rasterization	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
51	Tile-Based Triangle Rasterization	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
52	Edge-Based Triangle Representation 1/2	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
53	Edge-Based Triangle Representation 2/2	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.

Page	Slide title	How to read it
54	Rasterization Procedure	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
55	between adjacent triangles	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
56	Literature and other sources used in this chapter	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
57	Practice, Addison-Wesley.	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
58	Slide for Future: Sustainability in CG (@EG25)	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
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06.1 Line Rasterization

Source location: Section 6.1

Commentary

A mathematical line has infinitely many points, but a raster display has a finite grid. Line rasterization decides which grid cells or samples best approximate the ideal line. The algorithm must balance accuracy, speed, and consistency.

Incremental algorithms avoid recomputing expensive formulas for every pixel. The important idea is to step along one axis and update an error term that decides when to step along the other axis.

Mental model

Rasterizing a line means approximating a continuous path by grid decisions.

Check yourself

Why are incremental error updates useful in line rasterization?

06.2 Triangle Edge Rasterization

Source location: Section 6.2

Commentary

Triangles are the core primitive in real-time rendering. Edge functions or half-space tests decide whether a sample lies inside the triangle. This allows the rasterizer to test coverage systematically.

Coverage is not final visibility. A covered sample becomes a fragment candidate. Depth testing, stencil testing, blending, and other operations still decide whether the framebuffer changes.

Mental model

Triangle rasterization answers the question: which samples are inside this projected triangle?

Check yourself

What later stage can reject a fragment after triangle coverage succeeds?

06.3 Region Filling

Source location: Section 6.3

Commentary

Region filling generalizes the idea of determining interior samples. The algorithm must identify connected areas or spans that should receive color. This connects rasterization to scan conversion and image-space reasoning.

The practical issue is avoiding gaps, double fills, or inconsistent boundary handling. Small choices at edges can become visible artifacts when many primitives meet.

Mental model

Filling is about turning boundary descriptions into consistent interior samples.

Check yourself

Why can inconsistent edge rules create cracks between adjacent primitives?

06.4 Scanline-Based Triangle Rasterization

Source location: Section 6.4

Commentary

Scanline rasterization processes horizontal rows of the image. For each row that crosses a triangle, the algorithm finds the covered interval and fills the span. Attribute interpolation can be updated along edges and across each span.

The method is intuitive because it follows the memory layout of many images. It also demonstrates why interpolation is tied to rasterization: once you know a sample location inside the primitive, you can interpolate depth, color, normals, or texture coordinates.

Mental model

Each scanline asks: where does this triangle enter and leave this row?

Check yourself

Which attributes might be interpolated while filling triangle spans?

06.5 Tile-Based Triangle Rasterization

Source location: Section 6.5

Commentary

Tile- or block-based approaches group pixels into small regions. This can improve locality and parallel work distribution. Modern GPUs often reason in blocks or tiles internally because rendering is massively parallel.

For the exam, connect this to efficiency. The mathematical goal is still coverage, but the implementation is organized to use hardware resources well.

Mental model

Block-based rasterization keeps the same coverage problem but changes the work organization.

Check yourself

Why is block organization attractive for parallel graphics hardware?

End-of-lecture summary

If you remember only one thing from Lecture 06, remember this: This lecture explains how continuous primitives become discrete fragments on a grid. Rasterization is the bridge from geometric descriptions to sample-based image generation.

Before moving on, you should be able to explain the lecture without reading slide bullets aloud. Use the vocabulary from the slides, but add causal links: why a step exists, what data it consumes, what it produces, and what can go wrong.